

ETX1100/ETX5900 Business Statistics
Week 1
Descriptive Statistics

Figure 1: ETX5900 source PDF, Week 1, page 1

Learning Outcomes:

1. Understand the unit structure, people and resources available

2. Extend your knowledge of Descriptive Statistics

3. Identify and visualise different data types using Microsoft Excel

Figure 2: ETX5900 source PDF, Week 1, page 2

Contact details

Chief Examiner: Charanjit Kaur

Workshop Lead: Charanjit Kaur (weeks 1-4), Alyssa Hu (Weeks 5-8)
Workshop Support: Miriam Marembo, Madhu Prasanna, Susama Mahnaz and Song Li

For all questions

- email to ***ETX5900.allcampuses-x@monash.edu***
- Please use your student email and refer to the email etiquette instructions when drafting an email

For questions on content

- Ask the chatbot, Post on the Q&A platform **or** Ask during consultations

Figure 3: ETX5900 source PDF, Week 1, page 3

Your Learning

The most important member of the teaching team is YOU!

Preparation BEFORE the workshop (Own Time):

- Watch weekly pre-recorded videos and review accompanying slides.

Live Workshops (Real Time):

- Attend and actively **engage** in your workshops. Otherwise, you will lose valuable marks as a result. Don't be late for workshops!

AFTER the workshop (Wrap Up):

- Review all material and complete the post-workshop quiz. You have two one-hour attempts. One week to complete.

Try to keep up to date – it is easy to fall behind, but challenging to catch up



Figure 4: ETX5900 source PDF, Week 1, page 4

Recommended Text

Basic Business Statistics: Concepts and Applications, Berenson et al, Australian 5th edition, Pearson, 2018, ISBN 9781488620201.

- This textbook is available online via the Monash Library here.
- An e-book can be purchased using the link here
- An e-book can be rented using the link here

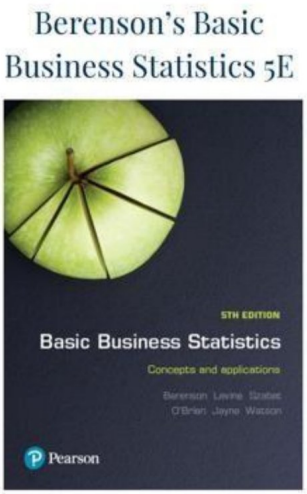


Figure 5: ETX5900 source PDF, Week 1, page 5

important

Assessment task	Value	Due date	Group or Individual	learning Outcomes Assessed
Workshop participation and in-class quiz	15% Best 6 out of 8	In-class	Individual	All
Take Home Quizzes	20% Best 6 out of 7	Quizzes **open every Tuesday at 11.55 pm after the workshop and dose at 11.55 pm on the following** Tuesday. • **2 attempts** for each Quiz • **60 minutes for each** attempt.	Individual	All
Assignment (Individual)	15%	19th of January 2025, 11:55 pm	Individual	1, 2, 3 and 6
Final exam	50%	TBC: 9-:ll3 Feb 2026	Individual	All

Figure 6: ETX5900 source PDF, Week 1, page 6

Learning Support



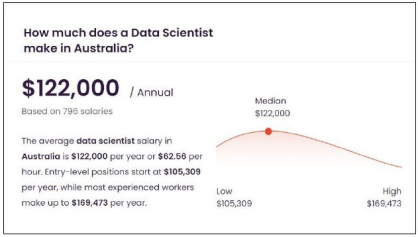
1. In-person and Zoom Consultations
2. **One update/reminder every Wednesday morning** (update for the previous week and preparation for the following week).
3. **Two personalised emails from the Chief Examiner** outlining your performance feedback (the first after Week 4 and the second after the assignment is marked).
4. List of **recommended textbook questions** every week. Solutions provided.

Figure 7: ETX5900 source PDF, Week 1, page 7

Build a Future in Data

Job Keyword	# jobs (17/07/2024)
“data”	24,721
“marketing” and “data”	5,052
“finance” and “data”	5,389
“human resources” and “data”	6,559
“logistics” and “data”	1,139
“business analytics”	2,555
“data science”	3,069

Source: seek.com



Job Keyword	Annual wages 17/7/2024
Accountant	$89,637
Marketing analyst	$100,980
Financial analyst	$110,756

Source: au.talent.com

Figure 8: ETX5900 source PDF, Week 1, page 8

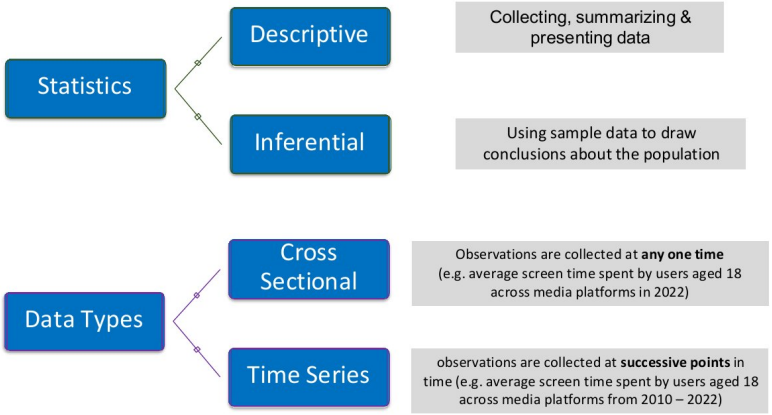
Descriptive Statistics

Textbook references: Berenson et al *Basic Business Statistics, 5th edition.*
Chapter 1: all
Chapter 2: 2.1, 2.2 (excluding stem and leaf plots), 2.3 (excluding polygons and cumulative percentage polygons), 2.4, 2.5, 2.6
Chapter 3: 3.1 (excluding the geometric mean), 3.2 (excluding the Chebyshev Rule), 3.4, 3.6

Learning Objectives:
Interpret business data using descriptive statistics techniques, including the use of Excel spreadsheet functions.

Figure 9: ETX5900 source PDF, Week 1, page 9

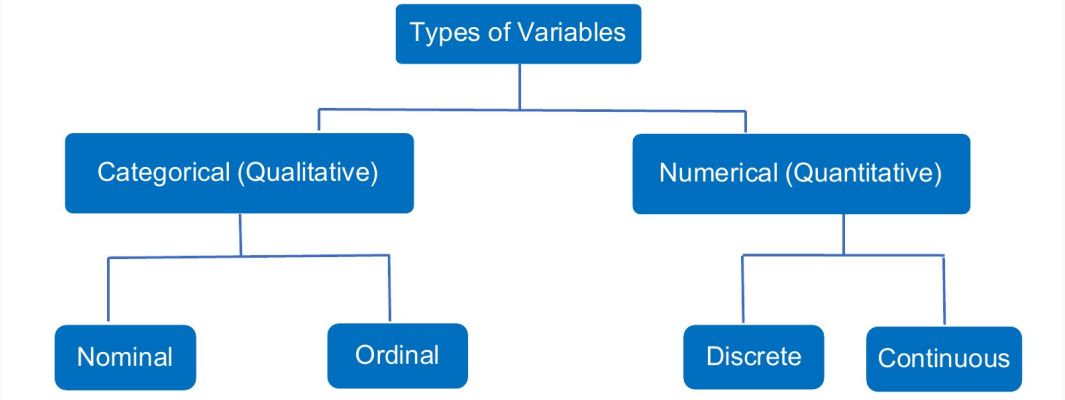
Branches of Statistics



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Figure 10: ETX5900 source PDF, Week 1, page 10

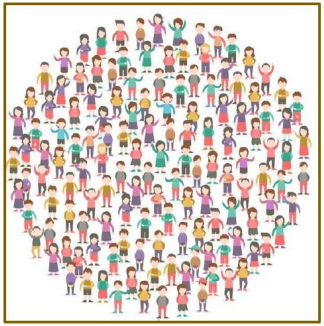
Types of Variables



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Figure 11: ETX5900 source PDF, Week 1, page 11

Basic Concepts: Population and Sample



Sample:
A subset of the population selected for analysis

- Often chosen *randomly*
- Preferably *representative* of the population

Statistic:
A numerical measure that describes a characteristic of a **s**ample

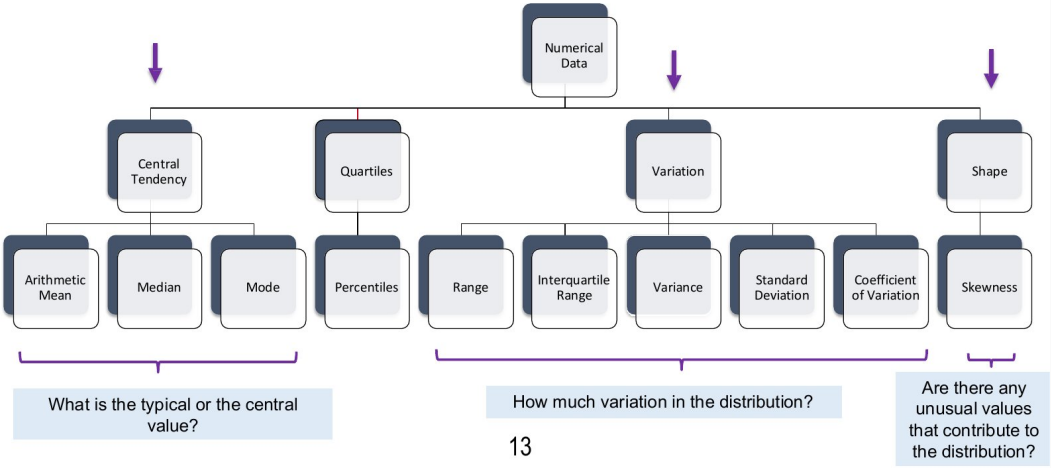


Population:
All members of a group about which you want to draw a conclusion.
(e.g. All voters in an election, all Telstra shareholders, all invoices submitted to Medicare for reimbursement, etc.)

Parameter:
A numerical measure that describes a characteristic of a **p**opulation.

Figure 12: ETX5900 source PDF, Week 1, page 12

Describing Numerical Data



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Figure 13: ETX5900 source PDF, Week 1, page 13

Variance and Standard Deviation

Variance: Mean of the squared deviations from the mean
Population variance: σ^2
Sample variance: s^2 (*Best estimate of population variance)*

- Disadvantage: Hard to interpret (the units are squared!)

Standard Deviation: Defined as the square root of variance ($\sqrt{Var}$)

Population standard deviation: σ
Sample standard deviation: s *(Best estimate of population std dev)*

- Advantage: Easier to interpret as units are the same as the original data

Figure 14: ETX5900 source PDF, Week 1, page 14

Variance and Standard Deviation - example

Otis Elevator Company uses 1-inch-diameter elevator rails.
There are two potential suppliers. To decide between the suppliers, the manager obtains a sample of 10 rails from each supplier.

Which supplier is preferred?

	A	B	C	D	E	F
1	**Diameters from two suppliers**					
2						
3	**Data**			**Summary measures**		
4	Supplier1	Supplier2			Supplier1	Supplier2
5	1.00	0.96		Mean	1	1
6	0.98	1.05		Median	1	1
7	1.02	1.00		Mode	1	1
8	1.01	0.97		Variance	0.000133	0.001200
9	1.00	1.00		Standard Deviation	0.0115	0.0346
10	0.99	1.03				
11	0.99	0.98		**Range names**		
12	1.00	1.02		Supplier1: A4:A14		
13	1.01	0.95		Supplier2: B4:B14		
14	1.00	1.04				

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Figure 15: ETX5900 source PDF, Week 1, page 15

Otis Elevator Company

Mean and median: 1 inch for both suppliers

Summary measures		
	Supplier1	Supplier2
Mean	1	1
Median	1	1
Mode	1	1
Variance	0.000133	0.001200
Standard Deviation	0.0115	0.0346

Variance: Supplier 2 > Supplier 1 (0.001200 > 0.000133)

- Supplier 1 preferred: less variation around the mean.
- However, variances are difficult to interpret because their units are *inches squared*, not *inches*.

Standard deviation: Supplier 2 > Supplier 1 (0.0346 > 0.0115)

- Supplier 1's standard deviation is about 1/3 of Supplier 2's standard deviation. Quite a disparity!

Given that Otis wants 1-inch diameter rails, Supplier 1 is preferred.

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Figure 16: ETX5900 source PDF, Week 1, page 16

Variance and Standard Deviation

Although each value in the data set is used in the calculation, there are several disadvantages:

- Sensitive to extreme values (outliers)
- Measures of absolute variation, not relative variation
- We cannot compare values that have different units

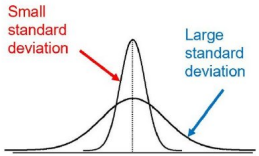


Figure 17: ETX5900 source PDF, Week 1, page 17

Coefficient of variation (CV)

- Measures **relative variation** (relative to the mean)
- Very useful for comparing the variation of two or more data sets, which are:
  - ✓ Measured in **different units** and/or
  - ✓ Are of **different magnitudes**

- The CV is ***always*** expressed as a percentage (%)

$$CV = \frac{Standard\ deviation}{Mean} \times 100\%$$

Figure 18: ETX5900 source PDF, Week 1, page 18

Coefficient of variation (CV)

Stock A

- Mean price last year: $50
  Standard deviation: $5

$CV_A = 5/50 * 100\% =$ **10%**

Stock B

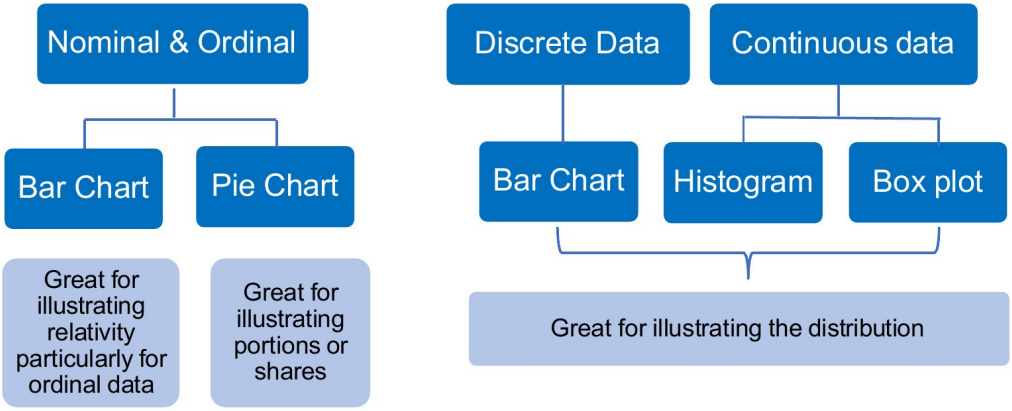
- Mean price last year: $100
  Standard deviation: $5

$CV_B = 5/100 * 100\% =$ **5%**

Same standard deviation, but stock B is less variable - relative to its mean.

Figure 19: ETX5900 source PDF, Week 1, page 19

Visualising Data



https://blog.hubspot.com/marketing/types-of-graphs-for-data-visualization
https://www.jmp.com/en_au/statistics-knowledge-portal/exploratory-data-analysis/types-of-graphs.html
https://piktochart.com/blog/types-of-graphs/
https://www.easel.ly/blog/types-of-graphs-and-charts-for-visualizing-data/

Figure 20: ETX5900 source PDF, Week 1, page 20

Getting ready with Excel: Data Analysis ToolPak

Windows:

Click the **File** tab → **Options** → **Add-Ins** category

In the **Manage** box, select **Excel Add-ins** → **Go**.

In the **Add-Ins** box, check the **Analysis ToolPak** check box → **OK**.

MAC

Click the **Tools** menu → **Excel Add-ins**.

In the **Add-Ins available** box, select the **Analysis ToolPak** check box → **OK**.

Figure 21: ETX5900 source PDF, Week 1, page 21

Having Fun with Data Types

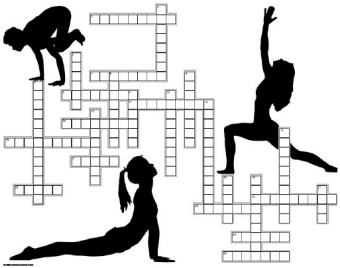
Form groups of three. In your groups, find **12 hidden words** in the 15 × 15 word search grid.

After finding the words, classify each one into the appropriate data type.
Use the four-column table provided below the crossword and fill in the words.

- Nominal Data (3 words)
- Ordinal Data (3 words)
- Discrete Data (3 words)
- Continuous Data (3 words)

You have **5 minutes** to complete this challenge!

Timer



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Figure 22: ETX5900 source PDF, Week 1, page 22

Task 1: Real Estate Data

We wish to compare the distribution of house prices for properties within/outside school zone

Refer to the workshop data set under Module 1. Refer to the Price column and answer the following questions.

a. *Generate summary statistics of house prices for properties within/outside the school zone*

b. *Draw two frequency percentage histograms.*

c. *Draw two box plots*

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Figure 23: ETX5900 source PDF, Week 1, page 23

Instructions for Summary Statistics

1. Click on Data tab and choose
2. Click on Data Analysis

3. Select Descriptive Statistics
4. Click ‘OK’

5. Highlight the relevant column and check ‘Labels in the first row’
6. Check ‘output range’ box and select a blank cell
7. Check the ‘Summary statistics’ box
8. Click OK

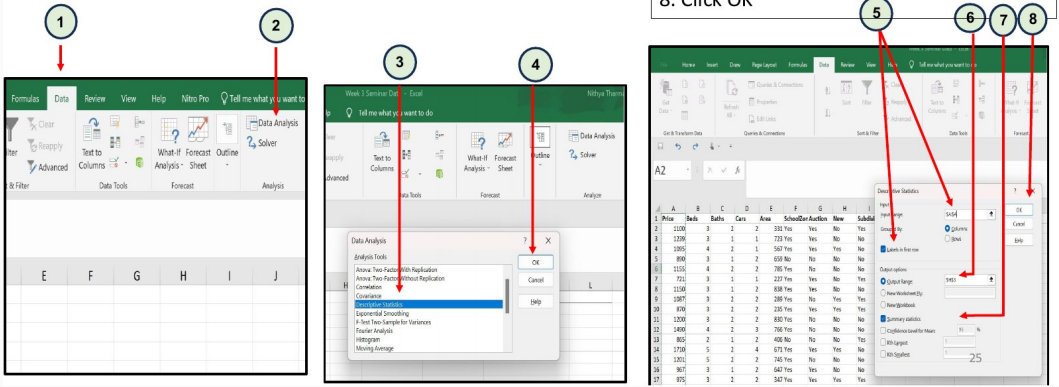


Figure 24: ETX5900 source PDF, Week 1, page 24

Average house prices

Let's compare the distribution of house prices within/outside school zone

	Price (School Zone)	*Price (Outside School Zone)*
Mean	1258.79	1046.04
Median	1225.50	980.00
Var	178200.41	130871.38
Stdev	422.14	361.76
Skewness	1.76	2.74
Q1	987.25	858.00
Q3	1405.25	1180.00
IQR	418.00	322.00
Range	3195.00	2430.00

Figure 25: ETX5900 source PDF, Week 1, page 25

Average house prices

Let's compare the distribution of house prices within/outside school zone

	Price (School Zone)	*Price (Outside School Zone)*
Mean	1258.79	1046.04
Median	1225.50	980.00
Var	178200.41	130871.38
Stdev	422.14	361.76
Skewness	1.76	2.74
Q1	987.25	858.00
Q3	1405.25	1180.00
IQR	418.00	322.00
Range	3195.00	2430.00

Houses in the school zone tend to be more expensive, with higher mean and median

Figure 26: ETX5900 source PDF, Week 1, page 26

Measures of Variability

- Range
- Inter-quartile range (IQR)
- Variance
- Standard deviation
- Coefficient of Variation (CV)

	Price (School Zone)	*Price (Outside School Zone)*
Mean	1258.79	1046.04
Median	1225.50	980.00
Var	178200.41	130871.38
Stdev	422.14	361.76
Skewness	1.76	2.74
Q1	987.25	858.00
Q3	1405.25	1180.00
IQR	418.00	322.00
Range	3195.00	2430.00

What do these statistics measure?

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Figure 27: ETX5900 source PDF, Week 1, page 27

Measures of Variability

Range → The difference between the maximum and the minimum values. It relies just on the two most extreme values in the dataset. **=MAX(…)-MIN(…)**

Interquartile Range (IQR) → the spread of the middle 50% of the data. IQR =Q3 -Q1

Q1 =QUARTILE.EXC(…,1) **Q3 =QUARTILE.EXC(…,3)**

Variance: average of the squared deviations (distance) from the mean **=VAR.S(…)**

Standard Deviation = $\sqrt{\text{Variance}}$. **=STDEV.S(…)**

Coefficient of Variation (cv): measures relative variation $= \frac{Standard\ deviation}{Mean}$ x 100%

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Figure 28: ETX5900 source PDF, Week 1, page 28

Variability of house price

	Price (School Zone)	*Price (Outside School Zone)*
Mean	1258.79	1046.04
Median	1225.50	980.00
Var	178200.41	130871.38
Stdev	422.14	361.76
Skewness	1.76	2.74
Q1	987.25	858.00
Q3	1405.25	1180.00
IQR	418.00	322.00
Range	3195.00	2430.00

There is a greater degree of absolute variability of houses in the school zone and only slightly lower relative variability.
Range: $3.195m vs $2.43m; IQR:$418k vs $322k; standard deviation: $422k vs $362k
CV: 33.54% vs 34.45% (standard deviation/mean *100%)

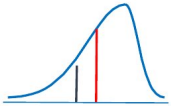
30

Figure 29: ETX5900 source PDF, Week 1, page 29

Shape of Distributions

Skewness measures how symmetric the distribution is
- Positive statistic → positive skewed, right tail longer than the left tail
- Negative statistic → negative skewed, left tail longer than the right tail

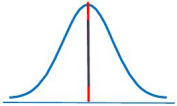
Left-Skewed (Skewness < 0)



Mean < Median

Some very "low" values
The mean is pulled left toward the "tail" of the distribution

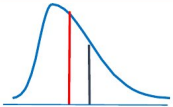
Symmetric (Skewness = 0)



Mean = Median

An equal number of "low" & "high" values. The mean stays in the centre of the distribution

Right-Skewed (Skewness > 0)

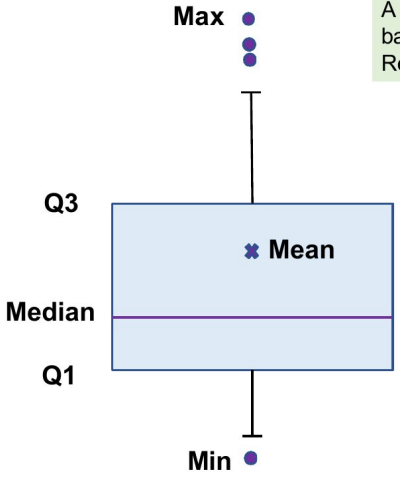


Mean > Median

Some very "high" values
The mean is pulled right toward the "tail" of the distribution

Figure 30: ETX5900 source PDF, Week 1, page 30

Visualization of Numerical Data using Box Plot



A graphical representation of the data based on the 5-number summary. Refer to page 121 of the textbook.

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Figure 31: ETX5900 source PDF, Week 1, page 31

Shape of Distributions and Box Plots

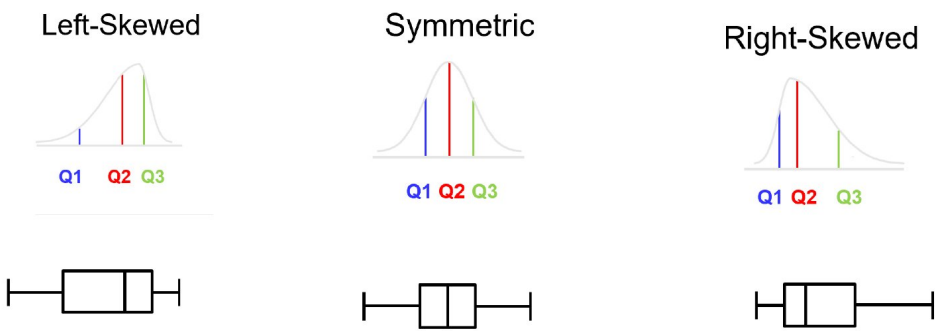


Figure 32: ETX5900 source PDF, Week 1, page 32

Example: Shape of the house price distribution

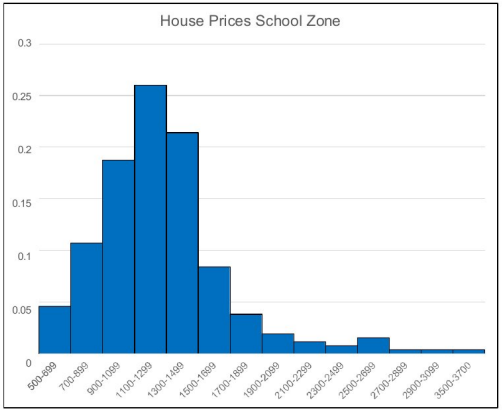
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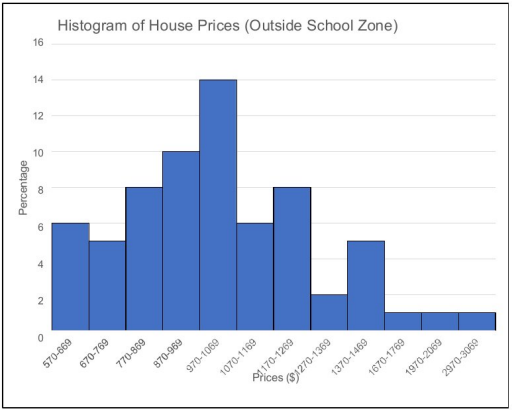
- Both distributions are positively skewed.
- There are some properties that are very highly priced

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Figure 33: ETX5900 source PDF, Week 1, page 33

Visualization of Numerical Data using Histograms





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Figure 34: ETX5900 source PDF, Week 1, page 34

Visualization of Numerical Data using Box Plot

Insert > Histogram >
Box and Whisker

You can plot multiple data series on the same plot
As long as they are of the same units!

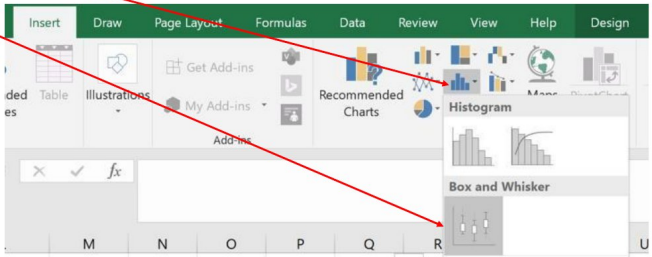


Figure 35: ETX5900 source PDF, Week 1, page 35

Visualization of Numerical Data using Box Plot



- Houses outside the school zone tend to be less expensive
- The distribution is less variable in absolute terms (tighter box) for houses outside the zone and only marginally higher (or almost the same) in terms of relative variability
- There are relatively more positive outliers for houses within the school zone.

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Figure 36: ETX5900 source PDF, Week 1, page 36

Task 2: Visualisation_Real Estate Data

Refer to the RealEstate worksheet

Variable	Type of Data	Type of graph
Price ($'000)	Numeric (continuous)	
No of bedrooms	Numeric (discrete)	

1. Construct relative frequency tables for the number *of bedrooms*
2. Draw an appropriate graph.

Figure 37: ETX5900 source PDF, Week 1, page 37

Solutions

Refer to chart

1. Construct a relative frequency table for the variable “No of Bedrooms”
2. Draw an appropriate graph.

Beds	Frequency	Relative Frequency
2	29	8.81%
3	138	41.95%
4	108	32.83%
5	45	13.68%
6	8	2.43%
8	1	0.30%
Grand Total	**329**	**100.00%**



Figure 38: ETX5900 source PDF, Week 1, page 38

Task 3: Visualisation for Continuous Data_Price

1. Construct a relative frequency table for the variable “Price”
2. Draw an appropriate graph.

Bin Range	Frequency	Relative Frequency
500-999		
1000-1499		
1500-1999		
2000-2499		
2500-2999		
3000-3499		
3500-3999		

Figure 39: ETX5900 source PDF, Week 1, page 39

1. Highlight one data cell
2. Insert ➔ Pivot table (in Mac Data ➔ Pivot table)
3. Select existing worksheet
4. Specify the cell location ➔ OK
5. Drag and drop price to the 'Rows' quadrant
6. Drag and drop price again to the 'values' quadrant

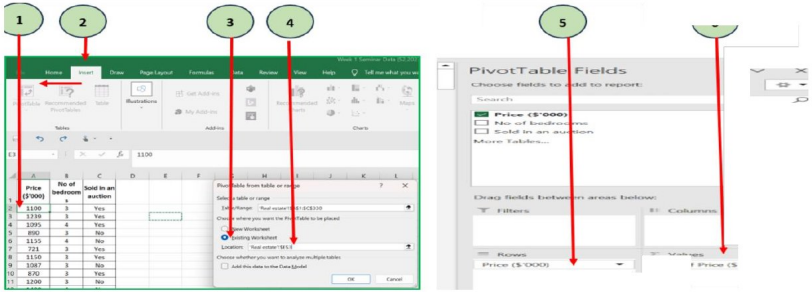


Figure 40: ETX5900 source PDF, Week 1, page 40

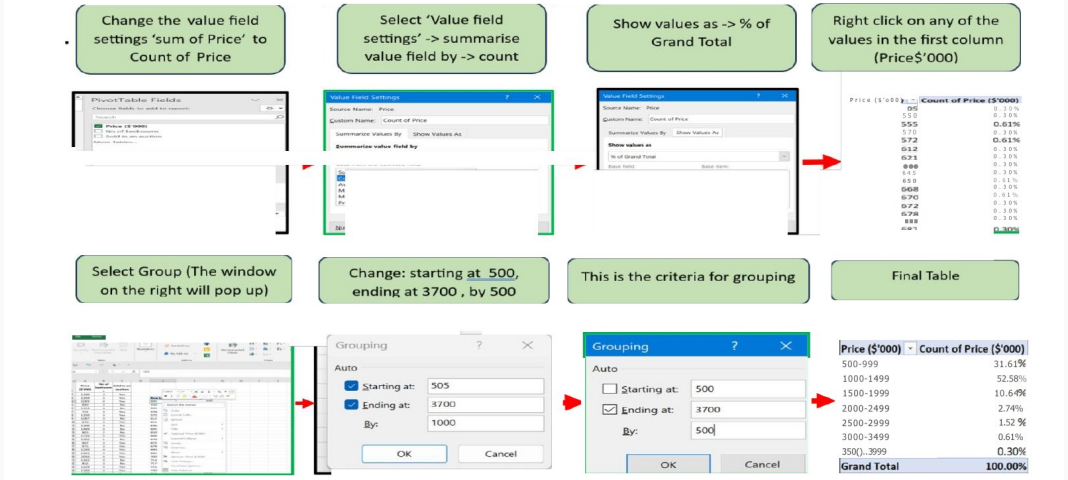


Figure 41: ETX5900 source PDF, Week 1, page 41

1. Highlight the pivot table
2. Insert a bar chart
3. To change the bar chart to a histogram, click on any of the bars in the bar chart
4. Right click and Select format data series
5. Change the gap width to O (zero)

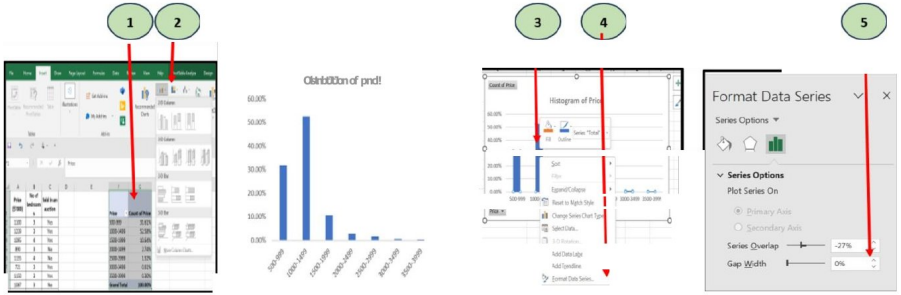


Figure 42: ETX5900 source PDF, Week 1, page 42

Solutions for Task 3

Bin Range	Frequency	Relative Frequency
500-999	104	31.61%
1000-1499	173	52.58%
1500-1999	35	10.64%
2000-2499	9	2.74%
2500-2999	5	1.52%
3000-3499	2	0.61%
3500-3999	1	0.30%

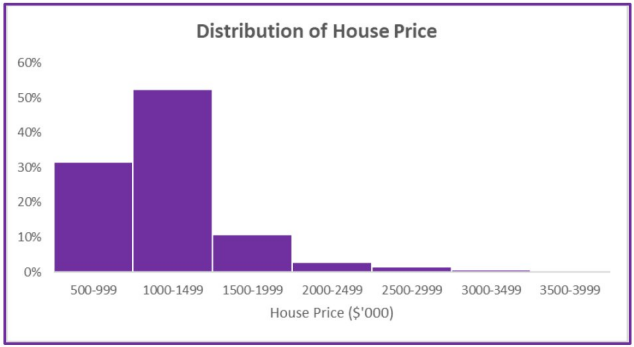


Figure 43: ETX5900 source PDF, Week 1, page 43

Key Takeaways

- Basic Statistical Concepts
- Data need not be numeric, it can be categorical and discrete
- Constructing frequency tables of data using Pivot Tables
- Charting different types of data

Figure 44: ETX5900 source PDF, Week 1, page 44